

Skin Cancer Classification - Project Report

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1. Introduction

The purpose of this project is to use two distinct convolutional neural network architectures to classify given images into 7 classes, representing 7 different types of skin cancers.

This project uses pre-trained deep learning models to classify skin cancer images into seven categories. The models employed are MobileNetV2 and MobileNetV3-Large, trained using PyTorch with data augmentation, class balancing techniques, and weighted loss functions.

2. Dataset and Distribution Irregularities

```
print("Training Images:")
for c1 in os.listdir(f"{TGT_DS_PATH}/train"):
    print(f"{c1}: {len(os.listdir(f'{TGT_DS_PATH}/train/{c1}'))}")
print()
print("Validation Images:")
for c1 in os.listdir(f"{TGT_DS_PATH}/val"):
    print(f"{c1}: {len(os.listdir(f'{TGT_DS_PATH}/val/{c1}'))}")
```

Training Images:

2:	148
7:	35
1:	112
5:	253
3:	343
4:	36
6:	2073

Validation Images:

2:	60
7:	20
1:	36
5:	97
3:	102
4:	15
6:	670

As can be seen from the image given above the dataset provided is highly imbalanced. Some classes such as 'nv' and 'mel' have a significantly higher number of samples compared to under-represented classes like 'df' and 'vasc'. Given the extent of this imbalance, the skewed representation of data had severe adverse effects on the predictive capability of the model. For classes such as 'vasc' having only 36 training images, a standard CNN model would be unable to learn the representative features of the class, leading to images belonging to the class being misclassified. To counter these obstacles we tried some standard and non-standard data augmentation techniques which would make the distribution more balanced.

3. Data Preprocessing and Augmentation

Firstly, as a control run we ran the models after performing standard augmenting procedures on the data such as horizontal flipping, resizing and rotation, this helped us to understand the effect of the skewed distribution in the accuracy of each class.

Secondly, we tried random sampling on the overrepresented classes, so as to balance the dataset. However, this only served to decrease the representation power of the selected class while not increasing the model's capability to capture the representative features of the heavily under-represented classes.

Next we tried the weighted loss method in which mistakes on rare classes are penalized more which forces the model to pay more attention to the minority classes. This did improve our results to some extent. Along with this, augmentation was applied to under-represented

classes with multipliers so as to ensure proportionate representation using random horizontal flipping, rotation, and resized cropping, with a cap of 15x the original sample size. Over-represented classes were downsampled to maintain balanced distribution but not to an extent where it overrides the apparent higher likelihood of such classes. This method gave us the best results and weights of the models trained after this method were the ones saved.

4. Training Process

We used pretrained models MobileNetV2 and V3 large due to them being fast efficient models which can be trained on smaller datasets. They also have many salient features such as inverted residual blocks , depthwise convolutions, hard swish activations etc. We also tried training these models from scratch but that gave us comparatively worse results. Both models were trained using CrossEntropyLoss with class weights computed from sample frequency. We tried SGD optimizer but found better results with Adam optimizer over multiple epochs. Training and Validation accuracy were tracked across train and validation sets, simultaneously. Accuracy and loss history was stored so that relevant curves could be plotted so as to visualize the results later.

5. Evaluation and Results

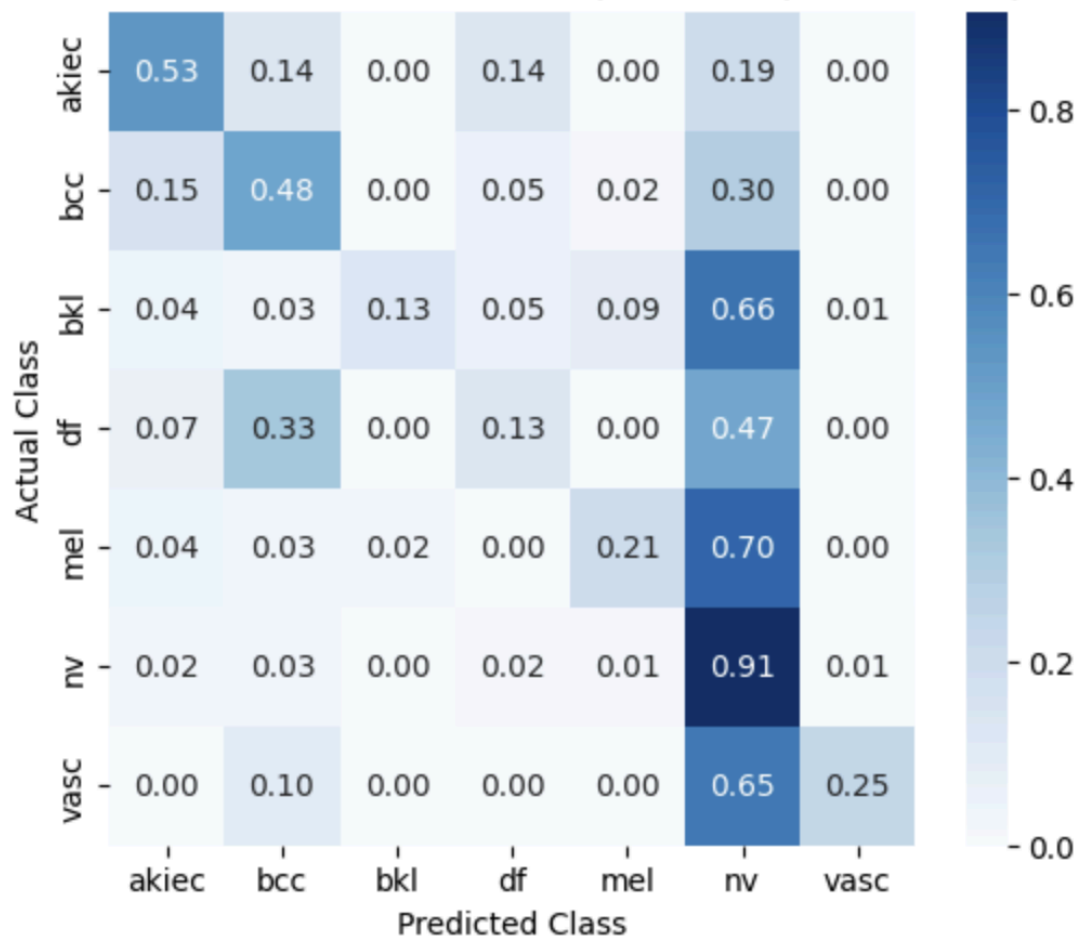
Model performance was evaluated using total accuracy, class-wise accuracy, and confusion matrices plotted via Seaborn. Over multiple runs MobileNetV3 generally outperformed MobileNetV2 in validation accuracy and class-wise consistency with MobileNetV3 achieving maximum validation accuracy of 74.90 % and MobileNetV2 achieving validation accuracy 72.40%. As expected, for both models the accuracy was highest for class 'vasc' the most

```
MobileNet V2 Validation Accuracy: 69.50%
Class 'akiec' accuracy: 52.78%
Class 'bcc' accuracy: 48.33%
Class 'bkl' accuracy: 12.75%
Class 'df' accuracy: 13.33%
Class 'mel' accuracy: 20.62%
Class 'nv' accuracy: 90.60%
Class 'vasc' accuracy: 25.00%

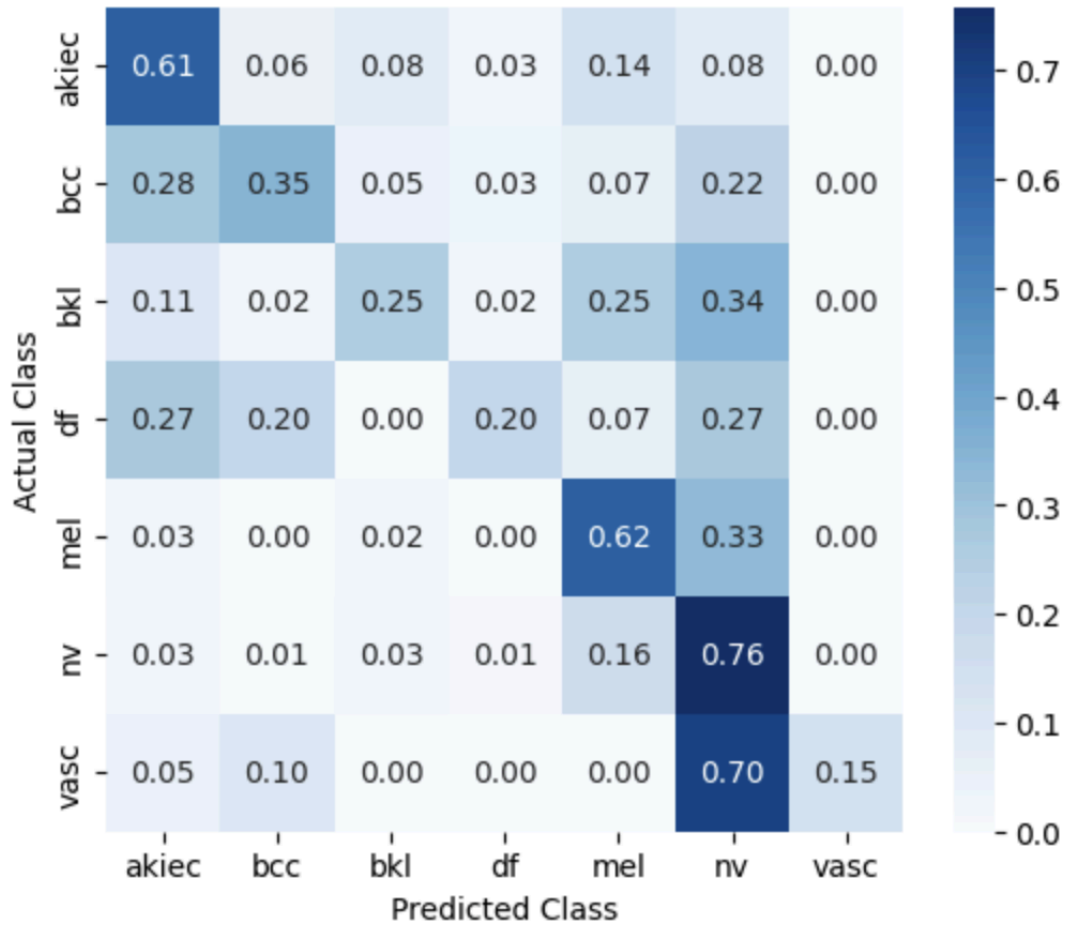
MobileNet V3-Large Validation Accuracy: 64.30%
Class 'akiec' accuracy: 61.11%
Class 'bcc' accuracy: 35.00%
Class 'bkl' accuracy: 25.49%
Class 'df' accuracy: 20.00%
Class 'mel' accuracy: 61.86%
Class 'nv' accuracy: 75.82%
Class 'vasc' accuracy: 15.00%
```

overrepresented class.

MobileNet V2 - Confusion Matrix (Validation, Normalized)



MobileNet V3 - Confusion Matrix (Validation, Normalized)



6. Final Notes

All models were trained on the rebalanced dataset and tested with consistent preprocessing steps. Final evaluation supports that effective balancing and augmentation significantly improved model generalization.